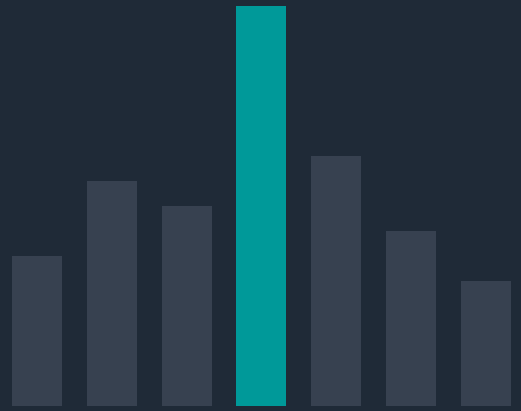
Rebuilding match physical insight after relegation

How I built a contextualised GPS pipeline to replace a lost league data feed



Michael Davies

Relegation removed the league data feed that match insight depended on

The performance question

- The league supplied match physical data in the division above; the new division does not
- The club’s legacy data ecosystem was lost with it
- The club had no systems or processes ready to replace it
- Practitioners and stakeholders still expected the same match insight

The question

How can practitioners and stakeholders keep receiving match physical insight, with context, without a league data feed?

Relegation cost a mature data ecosystem but left two sources to build on

What was lost

- Match-to-match comparison for the team and for each player
- Comparison with the opposition
- Season-long benchmarking against the rest of the league
- In- and out-of-possession context, which shapes how every number is read

What remained

GPS provider: a well-defined capture system, used mainly in training rather than matches

GPS provider API pipeline: already built for an earlier project

Broadcast tracking provider: league-wide tracking data, with speed zones that differ from the GPS provider’s

Two reports, each with its own purpose, replace the lost insight

I designed each report around the decisions its data can support

Contextualised GPS report

Question it answers

How hard did each player work, and in which phase of play?

Data

GPS provider data, split using match events from the event data provider

Ready

Within 2 hours of the match finishing

Used for

Player decisions, such as MD+1 training

Broadcast tracking provider trend report

Question it answers

Was the team higher than the opposition? How does it compare with the league?

Data

Broadcast tracking provider data, read at team level

Ready

24–36 hours after the match

Used for

Context only, not decisions

The broadcast tracking provider uses different speed zones, so it was used for team trends only

The speed zone problem

GPS provider

Speed zones set for the in-house GPS reporting

≠

Tracking provider

Its own speed-zone thresholds, applied across the league

→

Team trends

Compared by direction, not as absolute values

How the data was used

- Values for individual players were not reported
- Questions stayed at team level: was the team higher than the opposition, and where does it sit in the league?

Because the zone thresholds differ, broadcast tracking provider distances are not directly comparable with GPS provider outputs

A possession state model tags every moment of the match

I designed it to split play into in possession, out of possession and ball out of play

Event data provider match events
Event feed, processed straight after the game

→

Possession state function
Reads each event and assigns a state

→

State timeline
Dataframe tagging every moment of the match

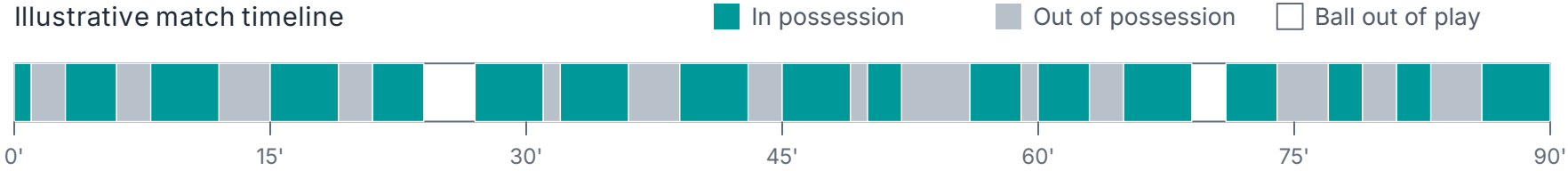
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Upload to provider
Timeline uploaded to the GPS provider through its API

→

Physical data by state
GPS provider data split by state and 15-minute block

Illustrative match timeline



Each player's physical output can be read for every 15-minute block and each possession state

The possession tags were validated against match video before staff relied on them

I lined up the feeds by timestamp and checked the tags on a sample of 5 matches

5 matches

assessed against match video before the tags were used in reports

Lining up the two feeds

The GPS provider and the event data provider both record Unix timestamps, so every GPS sample can be matched directly to the match event happening at the same moment, with no manual syncing.

Checking the tags

I validated the tagging on a sample of 5 matches, assessing the model’s possession states against the match video to confirm it was separating in possession, out of possession and ball out of play correctly.

An automated pipeline delivers the contextualised report within 2 hours of the match finishing

I automated every step after the practitioner downloads the match data

01

Collect

The practitioner collects the GPS units and downloads the match data

02

Trigger

The event data provider processes the match events straight after the game, which triggers a possession CSV

03

Combine

The GPS provider API pulls each player's data, split by possession state and 15-minute block

04

Interpret

The report lands on the practitioner's computer, ready for them to add context from the game

How I built it

AWS: S3 for match files, Postgres for querying

Containerised project: the same environment wherever it runs

Quarto: every report rendered as a club-branded PDF

The system was built in three layers, each adding context to the one before

I built and launched each layer in turn

Layer 1

Post-match GPS pipeline

A practitioner collection routine and an automated pipeline through the GPS provider API

Layer 2

Broadcast tracking provider benchmarking

Team-level trends against the opposition and the rest of the league

Layer 3

Possession context

In- and out-of-possession tags from the event data provider, plus 15-minute splits

Each layer went live and was trusted by staff before the next was added

The GPS report now arrives within 2 hours of the match finishing, not after a 10–12 hour wait

Last season

10–12 h

Waiting for the league's data to process before any report could go out

This season

Within 2 h

Collect GPS, download, shower and change: the report is ready to send

Broadcast tracking provider report: still 24–36 hours, which is acceptable because it adds context and is not used to make decisions

A 2-hour turnaround lets staff adjust MD+1 training using that match's data

An example of how staff use the report

The situation

A substitute comes on at 60 minutes. A player with 30 minutes would normally be expected to train on MD+1.

What the report shows

Within 2 hours of the match finishing, it shows a sprint load similar to what would be expected over 90 minutes, not 30.

The decision

They still train on MD+1, but the session is modified to reflect the load they actually carried.

Practitioners are part of the process

- The report arrives on their own computer, within 2 hours of the match finishing
- They add their context and insight from the game to the summary
- Stakeholders receive the data and the practitioner's judgement in one output



The club kept its match insight after relegation, and now gets it faster

- Build on the systems already in place
- Match each data source to the decisions it can support
- Add layers only once the base is reliable
- Keep practitioners in the loop

Data: GPS provider · event data provider · broadcast tracking provider

GPS informs decisions within 2 hours; tracking adds context

Appendix · post-match physical analysis: what staff can expect, and when

GPS provider report		Broadcast tracking provider report
Within 2 hours of the match finishing	Availability	Within 24–36 hours of the match finishing
Team values compared to norms	Graphical insights	Match-specific comparison to opponent
1st / 2nd half compared to norms		Game-on-game comparison to opponent
15-minute period breakdowns		Physical metrics compared to league
Game-on-game trends		Run types compared to league
Individual player breakdowns		
In and out of possession breakdowns		
Relative to minutes played	Relative values	Relative to time spent in possession
Total distance, metreage, HS / VHS / sprint / >90% distance, sprint / >90% efforts, peak speed, accelerations and decelerations, explosive efforts	Metrics included	Total distance, VHS / sprint / HI distance, sprint efforts, run types, effective playing time, time in possession, minutes played
Focuses on team and player workload relative to norms and minutes played	Focus	Focuses on opponent, context, possession and league comparisons